STAT 408 Project Proposal

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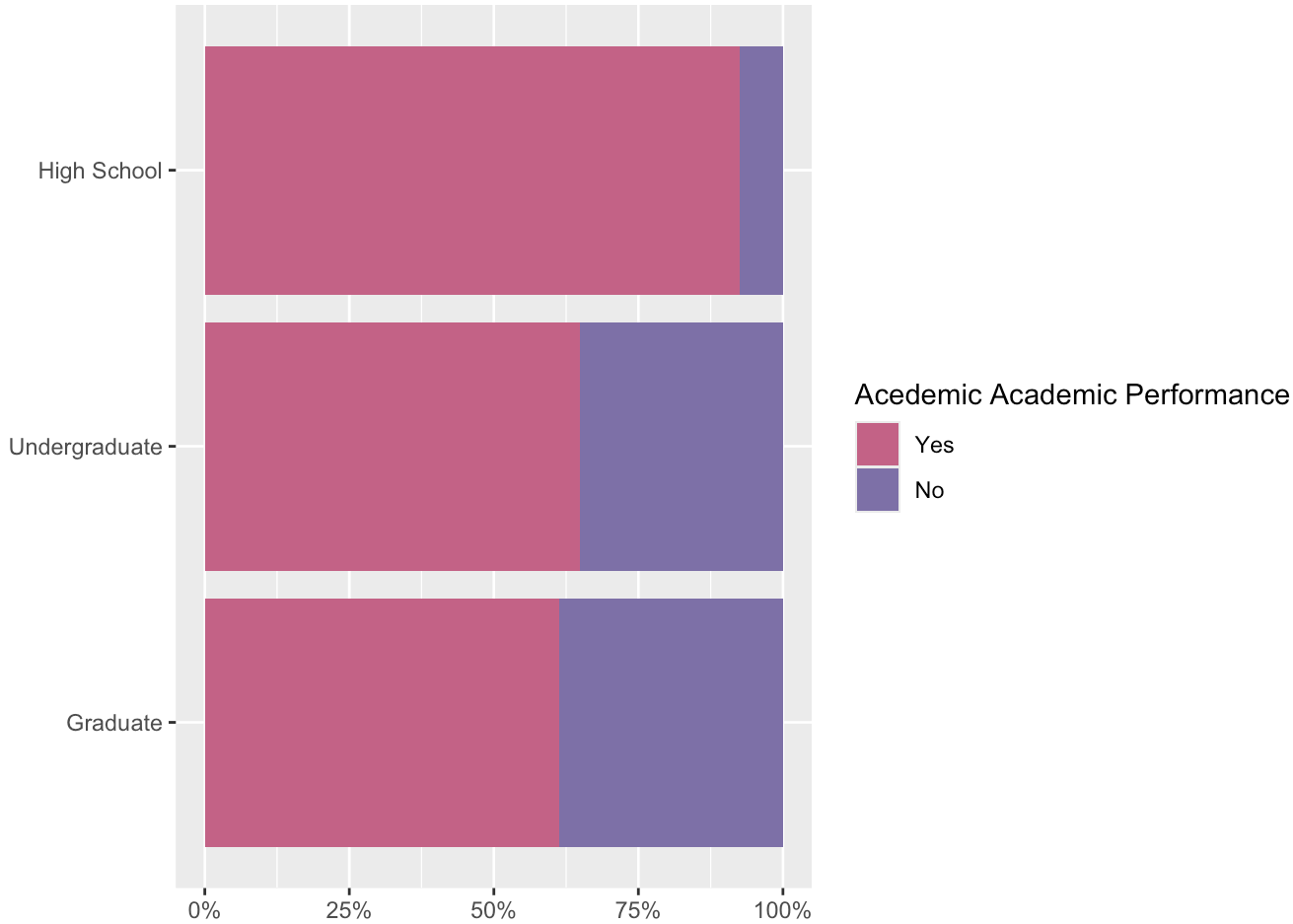
My project team is Lilly Price and Sierra Schall, and we are planning to work with Students Social Media Addiction. They come from a vast amount of countries with enrolled students between the age of 18 to 24. It represents the struggles that a lot students face with social media and academic performance worldwide. For our project, we are going to create an R dashboard, we are going to have overall effects, age, gender, and countries as our pages. For the first page, we are going to implement some tables and graphs containing variables that relate to academic performance, mental health, sleep hours, and addiction score. The second page,we are going to look at how age relates to social media addiction. We are going to be doing something similar within the gender and countries pages.

```
# Reading in the data
social <- read_csv("~/Desktop/Stat408/project proposal /Students Social Media Addiction.c

#Makes our data look a lot nicer and restricts it to the first five rows
social_table = social[1:5, ]

kable(social_table, format = "html") |>
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = F)
```

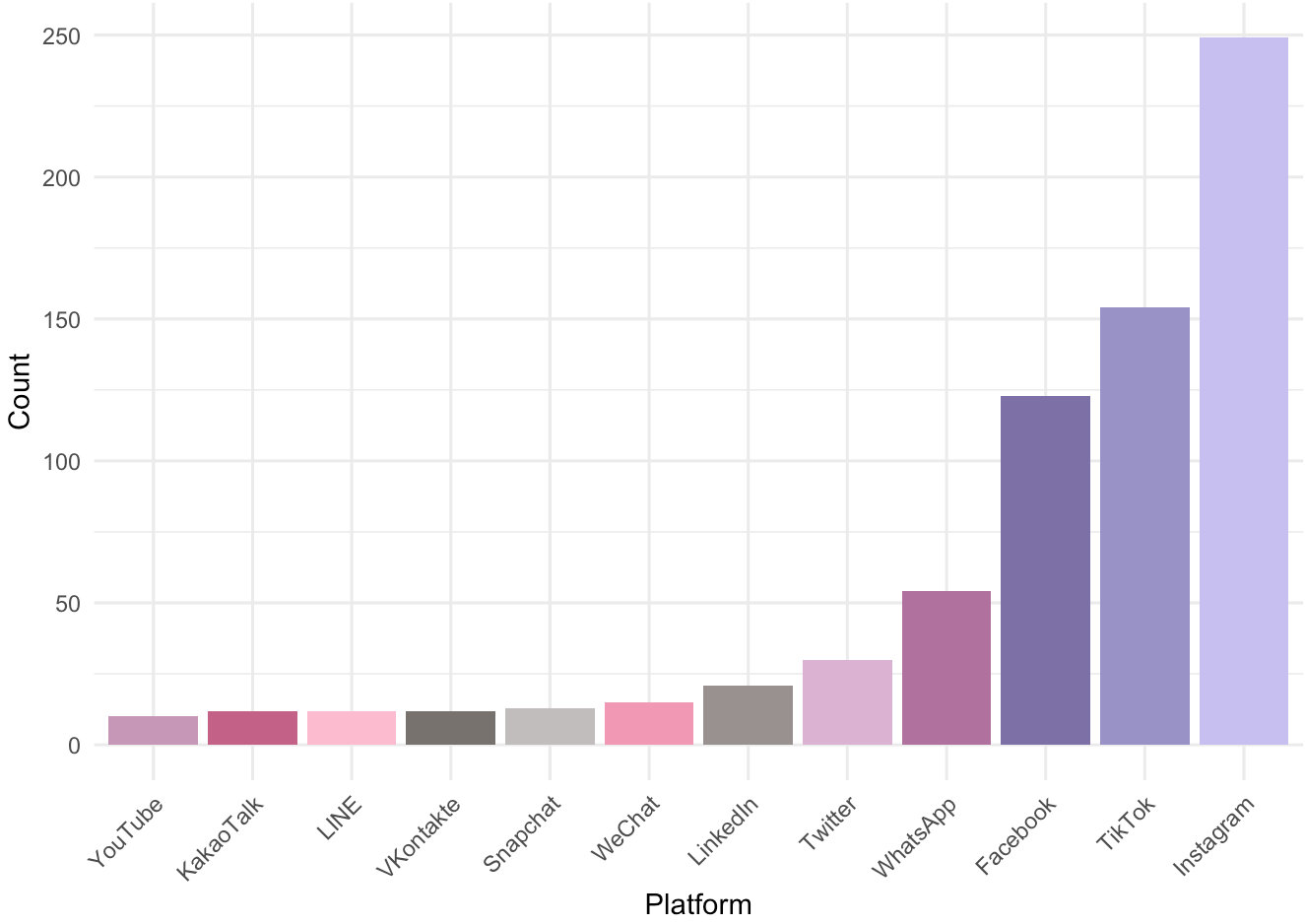
Student_ID	Age	Gender	Academic_Level	Country	Avg_Daily_Usage_Hours	Most_Used_Platform	Affec
1	19	Female	Undergraduate	Bangladesh	5.2	Instagram	Yes
2	22	Male	Graduate	India	2.1	Twitter	No
3	20	Female	Undergraduate	USA	6.0	TikTok	Yes
4	18	Male	High School	UK	3.0	YouTube	No
5	21	Male	Graduate	Canada	4.5	Facebook	Yes

```
#Creating levels and a factor function for my bar chart
social <- social |>
  mutate(
    Academic_Level = factor(Academic_Level,
                            levels = c("High School", "Undergraduate", "Graduate")))

#Creating a continuous bar chart
social |>
  count(Academic_Level, Affects_Academic_Performance) |>
  ggplot(aes(x = n, y = fct_rev(Academic_Level),
             fill = Affects_Academic_Performance)) +
  geom_col(position = "fill") +
  scale_fill_manual(
    values = c("#8074A8", "#C46487"),
    guide = guide_legend(reverse = TRUE)) +
```

```
  scale_x_continuous(labels = label_percent()) +
  labs(x = NULL, y = NULL, fill = "Acedemic Academic Performance")
```




```
#Creating a bar chart depicting how often the platforms are used amongst respondents
social |>
  count(Most_Used_Platform) |>
  mutate(p = n / sum(n)) |>
  ggplot(aes(x = reorder(Most_Used_Platform, n),
             y = n,
             fill = Most_Used_Platform)) +
  geom_col() +
  scale_fill_paletteer_d("ggthemes::Purple_Pink_Gray") +
  labs(x = "Platform", y = "Count") +
  theme_minimal() +
  theme(legend.position = "none",
        axis.text.x = element_text(angle = 45, hjust = 1))
```


250
200
150
100
50
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Count
YouTube
KakaoTalk
LINE
VKontakte
Snapchat
WeChat
LinkedIn
Twitter
WhatsApp
Facebook
TikTok
Instagram
Platform